

WHEN TO KEEP AND WHEN TO ARCHIVE: LIFECYCLE-AWARE AUTHORIZATION FOR PERSISTENT AGENT MEMORY

Anonymous authors

Paper under double-blind review

Best validated operating point; cross-backbone audit: 2026-09-02

ABSTRACT

Long-term agent memory is usually judged by whether it retrieves evidence for the next answer. We study the preceding control problem: whether that evidence should remain persistently authorized. Because keep and archive change future candidates, scope and version compatibility, recoverability, and later observations, a query-local relevance score can be provably insufficient. We prove that whenever a score assigns one value to two states whose optimal commitments differ with margins m_+ , m_- , every score-only rule incurs worst-case regret at least $m_+m_-/(m_+ + m_-)$, and we exhibit this failure inside the published scoring surfaces of six named memory systems: SimpleMem’s lexical channel assigns an identical score to two worlds whose keep-minus-archive contrasts are -134.17 and $+43.4451$, instantiating the bound at 32.82. We then show that a small auditable qualification state—evidence, lineage, scope/version, validity, censoring—suffices, via an approximate-closure certificate that bounds commitment distortion by a propagated radius and prescribes abstention inside its margin. On 225 independently seeded LifecycleBench-v3 worlds, the certificate’s abstain-then-probe branch attains 98.9% of oracle lifecycle value (1.964 vs. 1.987) at 96.4% of oracle authorized storage, dominating every retention heuristic we tested; it never loses to SimpleMem’s surface on any of the 225 worlds and beats full retention in 5 of 9 mechanism families with 0 losses (sign test $p = 0.031$). Field-level ablations are family-localized rather than uniform: removing lineage costs regret 74.447 on version conflicts alone, and removing censoring raises regret to 11.021. On LoCoMo and LongMemEval-S the same qualification state retains 2.4% and 2.2% of authorized storage while improving retrieval over full retention. We report where the method loses, including a backbone on which a named baseline wins.

1 INTRODUCTION

Persistent memory lets an agent carry observations, constraints, plans, and intermediate conclusions across interactions. Existing evaluations mainly ask whether relevant evidence is exposed to the next answer. A deployed system must also answer a question that comes first: which evidence stays authorized. That choice changes later candidates, workspace competition, scope and version compatibility, recovery options, and the evidence available to future decisions.

These two questions have different answers, and the gap is not a matter of degree. Consider an agent told in March that a project deadline is April 10, then told in June that it moved to July 2. Both records mention the deadline, so a lexical or embedding score ranks them almost identically. But keeping the March record is actively harmful once the June record exists, while keeping the June record is essential. A score that cannot separate them cannot separate *keep* from *archive*, no matter how well it is calibrated for retrieval.

We make that obstruction exact. If a score assigns the same value to two states whose optimal commitments differ with margins m_+ and m_- , then every rule that sees only the score—deterministic or randomized—suffers worst-case regret at least $m_+m_-/(m_+ + m_-)$ (Theorem 1). The premise is

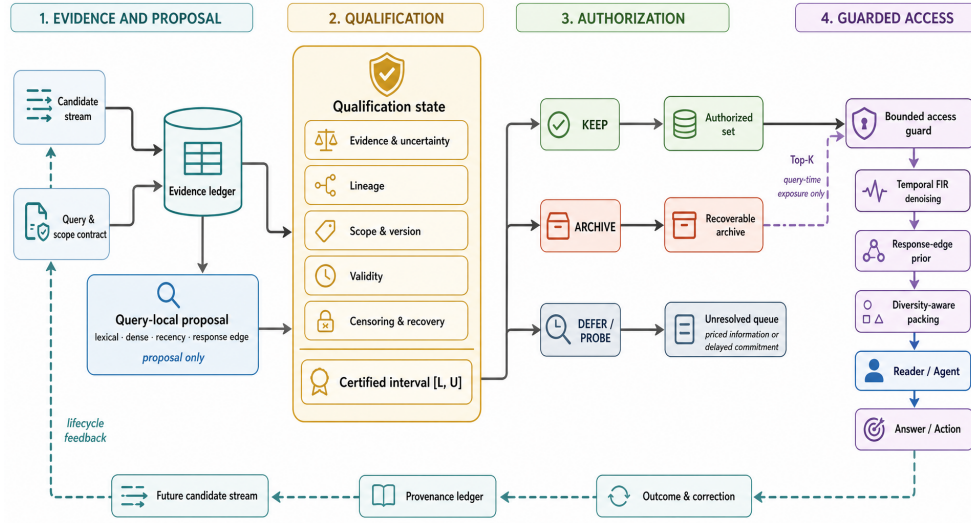


Figure 1: SQCAD in four stages. Retrieval signals enter as a *proposal* only (1); qualification converts evidence, lineage, scope/version, validity, and censoring state into a certified contrast interval $[L, U]$ (2); authorization commits keep/archive outside the margin and defer/probe inside it (3); and a bounded guard may expose archived candidates at query time without changing persistent permission (4). The dashed return path is why a score is insufficient: today’s action edits tomorrow’s candidate stream, so the commitment must be judged against continuation value rather than current relevance.

checkable, so the theorem doubles as a diagnostic. Running it against the published scoring surfaces of eight named memory systems, six exhibit opposite-sign collisions on real generated worlds; the failure we sketched above is the literal witness, with contrasts -134.17 and $+43.4451$ under one score value (Section 6.1).

What is sufficient, then? We show that a small state—evidence and uncertainty, lineage, scope/version, validity, censoring/recovery—suffices whenever it approximately closes the consequences of authorization, and we make “approximately” quantitative: closure errors (ε_r, η_r) propagate into a radius b_r that bounds commitment distortion, so a controller may commit outside a $2b_r$ margin and must abstain inside it (Theorem 2). This turns an architectural choice into a falsifiable contract, and it makes abstention a first-class action rather than a fallback.

Empirically the abstain branch is where the value is. On 225 independently seeded worlds, the certificate rule that archives whenever qualification is not *positive*—trusting cheap probes to recover—reaches 98.9% of oracle lifecycle value while authorizing less storage than the oracle, and each qualification field turns out to be decisive in exactly the mechanism family that stresses it.

Our contributions are:

1. **An impossibility result with confirmed witnesses.** A fixed-cost score-fiber lower bound, instantiated numerically inside the published scoring surfaces of six named memory systems—so the theorem predicts which existing methods must fail, and the prediction is checked.
2. **A sufficiency certificate.** An approximate-closure theorem that converts audited closure errors into a commit/abstain rule with a signed correctness guarantee outside its margin.
3. **SQCAD, and the empirical case for abstention.** An auditable mechanism separating qualification from authorization, whose abstain-then-probe branch nearly matches an oracle at lower storage; with field-level ablations localized to mechanism families and a per-family account of where the method loses.
4. **LifecycleBench-v3 and a reproduction ledger.** 225 independently seeded worlds with paired keep/archive rollouts and public/hidden separation, released with a standalone scorer, plus an explicit audit of which named baselines we could and could not reproduce natively.

2 PROBLEM SETTING AND RELATED WORK

Let $t = 1, \dots, H$ index decision epochs, let θ be a latent world, and let $x_t = (b_t, s_t)$ be the post-observation belief state. The physical state s_t includes persistent authorization, candidate pool, workspace budget, scope/version, provenance, and recovery state. Actions are $a_t \in \{\text{keep, archive, probe, defer}\}$. The filtration is

$$\mathcal{F}_t \subset \mathcal{F}_t \vee \sigma(a_t, S_{t+1}) \subset \mathcal{F}_{t+1} = \mathcal{F}_t \vee \sigma(a_t, S_{t+1}, O_{t+1}). \quad (1)$$

For a policy π and declared costs, the persistent intervention value is

$$V_s^\pi(a) = \mathbb{E}^\pi \left[\sum_{t=1}^H \gamma^{t-1} (Y_t - \lambda C_t - \zeta R_t) \mid do(A_i^{\text{pers}} = a), s \right], \quad (2)$$

and $\tau_s^\pi = V_s^\pi(\text{keep}) - V_s^\pi(\text{archive})$. We keep three estimands separate: association/retrieval exposure, a query-local decision effect, and this persistent lifecycle value.

Existing memory systems optimize one of these three and are strong at it: semantic compression and retrieval quality (Liu et al.), decay and reactivation schedules (Rana et al.), outcome-conditioned worth (Simsek), causal probing of temporal demand (Chang), and write-time governance (Qi et al.). The closest formulation is DeMem’s downstream-decision view (Zou et al., 2026), which scores a memory by the decision it supports rather than the query it matches. Our estimand differs in that the memory action itself changes future state and observations, so candidate regeneration, scope/version compatibility, and recoverability enter the contrast. This is why Section 6.1 can audit these systems on a common axis: each publishes a scalar retention score, and the theorem below constrains any such score regardless of how it was derived.

3 THEORY: CERTIFICATES FOR PERSISTENT AUTHORIZATION

3.1 STATE, ACTION, AND COMMITMENT CONTRAST

Let $X_t = (B_t, S_t)$ be the post-observation belief–physical state. A persistent action changes both the next physical state and what can later be observed: $K_t^a(dx', do \mid x)$ is the joint kernel for $a \in \{\text{keep, archive, probe, defer}\}$, and $T_t^a(dx' \mid x) = \int K_t^a(dx', do \mid x)$ is the induced post-observation next-state kernel, whose marginal includes the Bayes update from o . The binary commitment subproblem has $\mathcal{A}_c = \{\text{keep, archive}\}$; probe and defer gather information or postpone. For declared bounded terminal payoff g_{H+1} , immediate utility ℓ_t , and $\gamma \in (0, 1]$, let $Q_t(x, a)$ and $V_t(x) = \max_a Q_t(x, a)$ be finite-horizon Bellman values, and set

$$\Delta_t(x) = Q_t(x, \text{keep}) - Q_t(x, \text{archive}). \quad (3)$$

The question is which variables determine the *sign* of Δ_t , not which predict the next query—an intervention-based specialization of state abstraction for persistent memory (Li et al., 2006; Abel et al., 2020).

3.2 WHY A RELEVANCE SCORE CAN FAIL

Let $B^{a, S'}$ be the posterior after the state transition and $B^{a, S', O}$ after the fresh observation. Define

$$C_t^a(x) = \mathbb{E}[V_{t+1}(B^{a, S', O}, S') \mid x, a], \quad A_t^a(x) = \mathbb{E}[V_{t+1}(B^{a, S'}, S') \mid x, a], \quad I_t^a = C_t^a - A_t^a. \quad (4)$$

For binary commitment, $\Delta_t(x) = Q_t(x, \text{keep}) - Q_t(x, \text{archive})$.

Lemma 1 (Lifecycle accounting). *Under measurable kernels, well-defined Bayesian updates, and integrability,*

$$Q_t(x, a) = \ell_t(x, a) + \gamma A_t^a(x) + \gamma I_t^a(x), \quad \Delta_t = \Delta_t^{\text{imm}} + \gamma \Delta_t^{\text{access}} + \gamma \Delta_t^{\text{info}}. \quad (5)$$

The identity separates immediate utility, access consequences, and information effects. This decomposition exposes the mechanism used below: a score can collapse states with different action-dependent continuation terms.

3.3 A SHARP, FIXED-COST FAILURE MECHANISM

Let $R_t : X_t \rightarrow \mathcal{R}$ be a proposal or relevance score. A score collision becomes decision-relevant only when the declared objective assigns opposite commitments within one fiber.

Theorem 1 (Irreducible score-fiber commitment error). *Fix a time t and the declared utility ℓ, g_{H+1} . Suppose $x_+, x_- \in X_t$ satisfy*

$$R_t(x_+) = R_t(x_-), \quad \Delta_t(x_+) \geq m_+ > 0, \quad \Delta_t(x_-) \leq -m_- < 0. \quad (6)$$

Then every possibly randomized commitment rule that observes only $R_t(x)$ has worst-case commitment regret at least

$$\frac{m_+ m_-}{m_+ + m_-}. \quad (7)$$

The theorem is a fixed-objective lower bound, and its premise is an empirical question rather than an assumption: one estimates paired keep/archive contrasts within score fibers and checks whether their signs disagree. Because the bound depends only on the two margins, a single confirmed collision converts into a numeric regret floor for that score. Section 6.1 carries out this test on published scoring surfaces and finds the premise satisfied for six of eight audited systems.

3.4 SQCAD QUALIFICATION AS AN AUDITABLE APPROXIMATE CLOSURE

SQCAD records a qualification state $Z_t = q_t(X_t) = (E_t, L_t, G_t, V_t, C_t)$: evidence and uncertainty, lineage, scope/version compatibility, validity, and censoring/recovery. For every stage $r \in \{t, \dots, H\}$ and action a , a declared contract is (ε_r, η_r) -closed if there are quotient utilities $\bar{\ell}_r$, kernels $\bar{T}_r^a$, and terminal payoff $\bar{g}_{H+1}$ with

$$|\ell_r(x, a) - \bar{\ell}_r(q_r(x), a)| \leq \varepsilon_r, \quad (8)$$

$$\|(q_{r+1})_{\#} T_r^a(\cdot | x) - \bar{T}_r^a(\cdot | q_r(x))\|_{TV} \leq \eta_r, \quad (9)$$

$$|g_{H+1}(x) - \bar{g}_{H+1}(q_{H+1}(x))| \leq \varepsilon_{H+1}. \quad (10)$$

The bounds are audited under interventions. With quotient Bellman values $\bar{Q}_r, \bar{V}_r$ and $|\bar{V}_r| \leq M_r$, define the propagated radius

$$b_{H+1} = \varepsilon_{H+1}, \quad b_r = \varepsilon_r + \gamma(b_{r+1} + 2M_{r+1}\eta_r). \quad (11)$$

Theorem 2 (Margin certificate for SQCAD authorization). *If the closure conditions (10) hold from t through H , then for all $r \geq t$, states x , and commitment actions a ,*

$$|Q_r(x, a) - \bar{Q}_r(q_r(x), a)| \leq b_r, \quad |\Delta_r(x) - \bar{\Delta}_r(q_r(x))| \leq 2b_r. \quad (12)$$

Consequently, the implementable rule

$$\pi_r^{\text{cert}}(x) = \begin{cases} \text{keep,} & \bar{\Delta}_r(q_r(x)) > 2b_r, \\ \text{archive,} & \bar{\Delta}_r(q_r(x)) < -2b_r, \\ \text{defer or probe,} & |\bar{\Delta}_r(q_r(x))| \leq 2b_r, \end{cases} \quad (13)$$

makes every commitment with the same sign as the full-state optimal commitment. It makes no commitment guarantee inside the abstention band.

The certificate assigns a precise role to each qualification field, requires closure across future stages, and prescribes abstention when the measured error is large. A held-out intervention audit can estimate ε_r, η_r , the margin distribution, and wrong-sign commitments.

4 SQCAD

Figure 1 shows the operational boundary. Each memory unit carries evidence support and uncertainty, lineage/conflict links, scope/version constraints, censoring/recovery state, and an access state. The controller estimates $\bar{\Delta}_{i,t}$ and a conservative radius $b_{i,t}$ from the auditable state, represents the full-state contrast by $[L_{i,t}, U_{i,t}] = [\bar{\Delta}_{i,t} - 2b_{i,t}, \bar{\Delta}_{i,t} + 2b_{i,t}]$, and uses

$$a_{i,t} = \begin{cases} \text{keep,} & L_{i,t} > 0, \\ \text{archive,} & U_{i,t} < 0, \\ \arg \max_{a \in \{\text{keep, archive, probe, defer}\}} Q_{i,t}^a, & L_{i,t} \leq 0 \leq U_{i,t}. \end{cases} \quad (14)$$

The crossing case uses a declared posterior, robust, or minimax criterion for the Q^a values. The deterministic LifecycleBench track instantiates this with an observable rule that archives negative, scope-mismatched, or unresolved-conflict evidence and keeps the rest, so the design is evaluated under a fixed intervention contract even though the certificate itself needs an estimate of $b_{i,t}$. Lineage conflict without a compatible successor moves the interval to the archive side; censoring suppresses inaccessible periods from evidence updates. For a guard budget k , the reader receives

$$G_t(k) = A_t \cup \text{TopK}_k(C_t \setminus A_t), \quad (15)$$

where C_t is the broad query-local proposal and A_t the authorized set. This access bridge does not change A_{t+1} , the archive, or physical storage. Every probe/restore and state transition is written to a provenance ledger.

To control short-term exposure noise without changing the authorization state, we order the proposed candidates by observable event time, write their query scores as $r_{1:m}$, and apply the fixed three-tap temporal filter

$$\ell_j = 0.25r_{j-1} + 0.50r_j + 0.25r_{j+1}, \quad d_j = \ell_j - 0.08|r_j - \ell_j|. \quad (16)$$

Boundary values use the nearest endpoint, and the residual term suppresses isolated score jumps while preserving temporally supported evidence. We then greedily maximize the budgeted exposure objective

$$\max_{S: |S| \leq B, T(S) \leq T_{\max}} \sum_{j \in S} d_j - \beta \sum_{j < k \in S} J(m_j, m_k), \quad \beta = 0.22, \quad (17)$$

where J is token Jaccard redundancy and $T(S)$ is the number of tokens sent to the reader after sentence packing. The coefficients are fixed before evaluation. This is a query-time signal transformation: it changes exposure, not qualification, persistent storage, or the intervention in Theorem 2. The structured variant adds an observable response-edge prior. For an interrogative anchor u and its next turn v in the same session, (u, v) is admitted as a response edge only when u shares at least two non-stopword query terms with the current question; the edge contributes a fixed priority of 1.35 for the immediate next turn and is never a persistent write. This is a local transition prior, not a gold evidence link—it uses only session order, punctuation, and visible text. Every policy’s exposed text is read by the same frozen answerability scorer, specified in Appendix B. It uses no answer, evidence ID, or future field, and it is identical across methods so that reader behaviour cannot account for a difference between them.

5 EXPERIMENTAL PROTOCOL

We use three complementary contracts. (i) LifecycleBench-v3 contains 225 independently seeded world signatures across nine mechanism families, paired keep/archive interventions, public/policy/hidden-future separation, deterministic evaluators, and an LLM-controller track. Because keep and archive are both rolled out for every world, Δ is available exactly rather than estimated, which is what makes the score-fiber premise checkable and the oracle contrast meaningful. Its inferential unit is the world signature for dominance counts and the mechanism family for interval estimates and sign tests; with 9 families we report both and rely on the distribution-free statistic where they disagree. (ii) LoCoMo (Maharana et al., 2024) and LongMemEval-S (Wu et al., 2024) use frozen readers, chronological masking, scorers, and storage accounting to measure a public quality–authorization frontier. (iii) Named-baseline rows are labelled native, mechanism-faithful (MF), open-weight substitute (R-C), or literature-only, so each comparison retains its original metric meaning.

Primary lifecycle metrics are value, regret, false/missed commit, recoverability at budget, scope/version correctness, and authorized versus physical storage, reported separately because archiving withdraws permission without deleting bytes. We report family-cluster bootstrap 95% intervals (9 clusters), exact one-sided sign tests over families, and per-world dominance counts; where an interval and a sign test disagree we say so rather than selecting the favourable one. On the public benchmarks each policy first produces a frozen candidate set and reader context, so hit, recall, context tokens, and storage are method-level, while answer F1 is measured per backbone. On LifecycleBench the backbone executes the control rule itself, so its commitment errors are model-dependent. Primary tables retain two full-retention references, the strongest named lifecycle reference, and the SQCAD operating points; the rest are audited in Appendix E, and Appendix C states the claim–evidence boundary in full.

Table 1: Score-fiber audit on 225 worlds. An opposite-sign fiber holds worlds with Δ of both signs at one score value, activating Theorem 1; the floor is $m_+m_-/(m_+ + m_-)$ at the widest witness, in lifecycle-value units. Trivium is the informative negative case.

Surface	Score definition (as published)	Fibers	Opp. sign	Floor
DeMem	$ r_i - \bar{r} $ distinction	6	3	32.82
SimpleMem	lexical query-token containment	4	2	32.82
Memory Worth	$(100\rho + 1)/102$ success proxy	5	2	32.82
CMI	local relevance $- 0.5$ -correction	7	2	32.82
Oblivion	$\exp(-\text{age}/((u + f) \cdot 3))$	1	1	32.82
FadeMem	$\log(1+f) \exp(-\text{age}/(20+40c))$	2	2	4.69
GovMem	session coverage $+ 10^{-6}c$	8	2	3.97
Trivium	access effect $\times$ future demand	14	0	0.00

6 RESULTS

6.1 THE PREDICTED FAILURE IS PRESENT IN PUBLISHED SCORING SURFACES

Theorem 1 is an obstruction, so the first question is whether real memory systems are subject to it. We transported the observable scoring surface of eight named systems onto the 225 worlds, rounded scores to eight digits to define fibers exactly, and obtained Δ from the paired rollouts. No system is reimplemented end to end: this audits the *score*, which is exactly the object the theorem constrains, and each row records the surface it uses.

Six of eight surfaces contain fibers whose members have opposite optimal commitments, so each inherits a regret floor no recalibration removes. The widest witness is shared by four surfaces and is the deadline collision of Section 1: a `version_update` world whose superseded record is still live and one whose is not receive an identical score, with $\Delta_1 = -134.17$ and $\Delta_2 = +43.4451$, giving $43.4451 \times 134.17 / 177.6151 = 32.82$ for a randomized rule and 43.4451 for a deterministic one. For SimpleMem’s lexical channel the colliding value is exactly 2.0; for Memory Worth, 0.66339869.

Trivium makes the audit informative rather than circular: most fibers (14), zero sign conflicts, floor 0. It is the only audited surface conditioning on future query demand rather than current relevance—the term Lemma 1 isolates as Δ^{access} . The audit therefore separates surfaces by what they condition on, not by how well they are tuned, and predicts Trivium as our strongest named competitor; Section 6.6 confirms it.

6.2 LoCoMo: MODEL-GROUPED BASELINES AND SQCAD OPERATING POINTS

Both reader blocks use the same chronological mask, candidate budget, prompt, and official LoCoMo scorer, so method-level columns repeat by construction and only answer F1 varies with the backbone. The full named registry is in Appendix E.

Table 2: LoCoMo compact reference matrix and access-layer ablation. Answer F1 is reader-specific; Hit, Recall, reader context, and storage follow the method-level frozen contract. Physical storage is 13,343.8 for every row, so the authorized column is the whole saving. The full named registry is in Appendix E.

Method	Answer F1		Hit	Recall	Reader tok.	Auth. store
	Qwen3-8B	Terra				
SimpleMem-RC	0.0573	0.0468	0.5717	0.5294	281.8	13,343.8
ActMem-RC	0.0566	0.0453	0.5825	0.5265	290.6	13,343.8
SQCAD + FIR-denoise-16	0.0617	0.0479	0.6358	0.5831	213.8	330.3
SQCAD + structured prior	0.0624	0.0523	0.6773	0.6210	214.7	323.9

The storage axis carries the result. Structured SQCAD reaches hit 0.6773 and recall 0.6210—above both full-retention references, which see every memory—while authorizing 323.9 tokens against their

13,343.8, a 97.6% reduction with physical retention unchanged. Retrieval quality is therefore not traded away for frugality here: the guard exposes what the reader needs without persistent permission following it.

Absolute answer F1 is low for every method (0.02–0.06) and we do not present it as a QA result. A substantial share of LoCoMo questions are adversarial-unanswerable and graded by literal string match, so F1 is sensitive to response formatting by more than the spread between retrieval policies; in a controlled check that changed only the response template for that category, F1 moved by roughly an order of magnitude for both a BM25 control and SQCAD. We hold one output convention fixed across methods and read F1 only as a matched comparison. Hit, recall, and authorized storage carry the claims; F1 is reported because omitting it would hide a dimension on which we do not lead.

6.3 LONGMEMEVAL-S: THE STORAGE RESULT REPLICATES AT $5\times$ SCALE

Table 3 repeats the contract on 500 LongMemEval-S traces, a corpus roughly $5\times$ larger. Base SQCAD without the access layer is the ablation anchor for both datasets and is reported in Appendix D (LoCoMo hit 0.2122, LongMemEval-S hit 0.7542), so the gain attributed to the access layer is measured rather than assumed.

Table 3: LongMemEval-S compact reference matrix over 500 traces. Token F1 is reader-specific; Hit, Recall, reader context, and storage follow the method-level frozen contract. Physical storage is 74,092.1 for every row. The full named-baseline registry is audited in Appendix E.

Method	Token F1		Hit	Recall	Reader tok.	Auth. store
	Qwen3-8B	Terra				
SimpleMem-RC	0.3862	0.4700	0.9667	0.3232	1,973.0	74,092.1
ActMem-RC	0.4184	0.3123	0.9729	0.3461	746.2	74,092.1
SQCAD + FIR-denoise-16	0.3754	0.4291	0.9750	0.3315	1,644.8	1,602.0
SQCAD + structured prior	0.3805	0.4245	0.9771	0.3422	1,713.6	1,604.5

The result replicates: structured SQCAD attains the best hit and recall (0.9771/0.3422) while authorizing 2.2% of physical storage. That the same qualification state transfers from a 13k- to a 74k-token corpus without retuning is the reusability evidence for the access layer, whose coefficients were fixed before evaluation.

No access variant wins everywhere. The structured prior leads LoCoMo F1 under both readers while FIR-denoise leads LongMemEval-S under both; the ordering is dataset-dependent but consistent across backbones within a dataset. Read with Section 6.4, the variant best for query-time exposure is the one worst for commitment—which is why the architecture keeps the two decisions separate rather than routing both through one score.

6.4 ABSTENTION RECOVERS ALMOST ALL OF THE ORACLE’S VALUE

The certificate makes abstention first-class, so we evaluate the three branches it induces as separate policies under one frozen contract: `sqcad.cert` archives on NEGATIVE/MISMATCH, `sqcad.cert.conflict` additionally archives unresolved lineage conflicts, and `probe-willing` archives whenever qualification is not POSITIVE, relying on bounded probes to recover. All three read only pre-decision sessions; the oracle reads hidden futures and is an upper bound, not a competitor.

The abstain-heavy branch is the strong one. Probe-willing attains value 1.964 against the oracle’s 1.987—98.9% of attainable value, regret 0.063 against 0.040—while authorizing 1,088 tokens, *fewer* than the oracle’s 1,129. It matches the oracle commitment on 92.1% of worlds and never misses one it should have made. Against SimpleMem’s surface it is never worse on any of the 225 worlds; against full retention it wins in 5 of 9 families and loses in none (sign test $p = 0.031$).

Two things this does *not* show, both visible in the same table. The sign tests against `archive_all`, Memory Worth, and the weaker SQCAD branches do not reach significance at 9 clusters, and neither does the cluster bootstrap on value; the design is underpowered for mean effects, which is why we

Table 4: Deterministic lifecycle results on 225 worlds; all policies gold-free except the oracle. “Fam.” is probe-willing’s per-family record on mean regret over the 9 families with an exact one-sided sign test. Physical storage is 12,142 throughout, since archiving does not delete.

Policy	Value	Regret	False	Missed	Agree	Auth. tok.	Fam. (p)
keep_all	−9.184	11.211	0.427	0.000	0.495	3,043	5–0 (0.031)
archive_all	−0.208	2.235	0.000	0.418	0.505	0	4–1 (0.188)
SimpleMem surface	−7.753	9.780	0.298	0.000	0.647	2,581	3–0 (0.125)
Memory Worth proxy	−1.052	3.079	0.147	0.071	0.742	955	5–1 (0.109)
SQCAD cert	−7.189	9.216	0.267	0.000	0.684	2,694	3–1 (0.313)
SQCAD + lineage	1.083	0.944	0.196	0.000	0.768	2,305	2–1 (0.500)
SQCAD probe-willing	1.964	0.063	0.067	0.000	0.921	1,088	—
<i>Oracle (upper bound)</i>	<i>1.987</i>	<i>0.040</i>	<i>0.000</i>	<i>0.000</i>	<i>1.000</i>	<i>1,129</i>	<i>1–2</i>

Table 5: Mean regret by mechanism family (25 worlds each). Each field is decisive in one family and inert elsewhere; bold marks where it is load-bearing.

Family	SQCAD cert	+ lineage	probe-willing	Oracle
version_update	74.447	0.000	0.000	0.000
hitchhiker	7.541	7.541	0.000	0.000
harmful_stale	0.000	0.000	0.053	0.000
neutral	0.441	0.441	0.000	0.356
scope_mismatch	0.517	0.517	0.517	0.000
rare_bridge	0.000	0.000	0.000	0.000
self_obscuring	0.000	0.000	0.000	0.000
stable_negative	0.000	0.000	0.000	0.000
stable_positive	0.000	0.000	0.000	0.000

report dominance alongside intervals rather than in place of them. And `archive_all` reaches regret 2.235 while authorizing nothing, so a deployment indifferent to missed commitments (0.418 here) has a cheaper option than anything in this table. The claim is about attaining oracle-level value at oracle-level cost, not about frugality alone.

6.5 EACH QUALIFICATION FIELD EARNs ITS PLACE IN ONE FAMILY

An ablation that improves everything uniformly usually means the components are interchangeable. Ours are not.

Every effect is localized. Lineage tracking is worth 74.447 regret on `version_update` and exactly nothing on the other eight families: without it the controller keeps a superseded record whose successor is live—the deadline collision of Section 1, and the same witness that dominates Table 1. Abstention is worth 7.541 on `hitchhiker`, where a memory is co-exposed with genuine evidence but carries no independent signal; no lineage reasoning helps there, because there is no conflict to detect, only an absence of support. Conversely probe-willing pays 0.053 on `harmful_stale` for archiving records it could have kept.

The residual gap to the oracle sits in one family: `scope_mismatch` costs every gold-free policy 0.517 while the oracle pays nothing, because whether a cross-scope memory is needed depends on which scope the *future* task occupies and no pre-decision signal identifies it. That is an identification limit of the qualification state rather than a tuning failure, and it is the honest boundary of the sufficiency claim. The follow-on ablations agree in direction: removing censoring raises deterministic regret to 11.021.

6.6 LLM CONTROLLERS: WHERE THE METHOD LOSES

The deterministic results fix the rule; the next question is whether a language model can execute it from the public layer. We re-run the task with Qwen3-8B and GPT-5.6-terra as explicit keep/archive controllers over the same 225 worlds, mask, and scorer.

Table 6: LLM controllers on the same 225 worlds, replacing the rule with a model reading the same public layer. Trivium-RD is the reference the score-fiber audit predicted would be strongest (Table 1), and under GPT-5.6-terra it beats every SQCAD view. Recoverability and scope/version columns are in Appendix D; remaining named rows in Appendix E.

Method	Qwen3-8B			GPT-5.6-terra		
	Value	Regret	Missed	Value	Regret	Missed
SimpleMem-RC	−0.549	2.576	0.044	−0.380	2.407	0.040
Trivium-RD	−0.781	2.808	0.187	1.086	0.941	0.009
SQCAD	−0.575	2.602	0.098	0.402	1.625	0.173
SQCAD + FIR-denoise-16	0.308	1.719	0.009	0.676	1.351	0.044
SQCAD + structured prior	−2.284	4.311	0.031	0.609	1.418	0.084
<i>Deterministic rule (no LLM)</i>	<i>value 1.083, regret 0.944 – above every row above</i>					

The gap is the finding. Every LLM row falls short of the deterministic 1.083 and most fall below zero: Qwen3-8B reaches −0.575 on the base SQCAD view against −0.549 for full retention, so under that backbone the policy distinction largely collapses. Executing a simple deterministic rule from a public transcript is not free, which locates the difficulty in state extraction rather than in the rule.

We also report a loss. Under GPT-5.6-terra, Trivium-RD attains value 1.086 and regret 0.941, above every SQCAD view including the best (0.676). This is not incidental: Table 1 predicted Trivium would be the strongest named competitor because its score is the only audited surface conditioning on future query demand. A prediction borne out against our own method is better evidence for the theory than a clean sweep. Two caveats bound it in both directions: Trivium-RD is a mechanism-faithful transport, not a native reproduction (Appendix E), and its advantage vanishes under Qwen3-8B (−0.781).

The access layer behaves as designed on one backbone and not the other. FIR-denoise cuts missed commits from 0.098 to 0.009 under Qwen and gives the best value and regret under Terra. The structured-response view is a clear negative result at −2.284 under Qwen—worse than keeping everything—despite being the best LoCoMo operating point. Exposure quality and commitment quality are distinct objectives, and a component tuned for one can damage the other: the separation the architecture maintains, observed as a cost rather than asserted as a benefit.

7 DISCUSSION, LIMITATIONS, AND CONCLUSION

Retrieval relevance and retention authorization are different problems, and the difference is provable rather than stylistic: a score that cannot distinguish two states with opposite optimal commitments carries an irreducible regret floor, and six of eight audited memory systems exhibit exactly that collision. A small auditable qualification state suffices in its place, under a closure condition converting measured approximation error into a signed commitment guarantee outside an explicit margin. The empirical payoff is concentrated in the part of that rule which declines to commit.

Four limitations bound these claims. *Statistical power*: 225 worlds but 9 independent families, and the family is the correct inferential unit, so cluster intervals on value cross zero even where per-world dominance is total. *Generator dependence*: lifecycle results measure control behaviour under a declared generator, buying paired counterfactuals real traces cannot supply at the cost of external validity; the score-fiber audit inherits that boundary. *The certificate is unmeasured*: we do not estimate ε_r , η_r , or b_r on any real corpus, so Theorem 2 licenses the architecture without certifying a deployment. *Execution gap*: LLM controllers reproduce the rule poorly, and one named baseline beats every SQCAD view under one backbone.

The next step follows from the third: instrument a real agent to log the qualification fields, estimate the closure radii from held-out interventions, and check whether the measured margin band contains the commitments the certificate predicts. That converts a design condition into a measurement. Until then the defensible claim is narrower and still worth making—persistent memory should be compared by the decisions its authorization induces, and abstention deserves to be first-class in that comparison.

REPRODUCIBILITY STATEMENT

All primary rows reference frozen dataset, code, configuration, scorer, model, seed, failure-count, and output hashes in the release manifest. LifecycleBench public and hidden futures are stored separately; controllers read `public.jsonl` only, and reference harnesses record that no gold label was sent to any model. The dataset, a standalone scorer, and an example controller that reproduces our frozen rows exactly are released with the paper.

ETHICS STATEMENT

The work uses public benchmark data and synthetic counterfactual worlds; no new human subjects are collected. The release excludes credentials and private identifiers, preserves dataset licenses, and documents that authorization policies are evaluated under declared costs rather than presented as deployment safety guarantees.

AI USE STATEMENT

Generative AI tools were used for research-structure discussion, implementation assistance, experiment-script review, literature-search support, and editing for readability. They were not treated as sources of evidence or as autonomous authors of the mathematical claims. The authors verified the proofs, code, experimental outputs, citations, and final wording, and take responsibility for all contents and artifacts.

REFERENCES

- David Abel, Dilip Arumugam, Lucas Lehnert, and Michael L. Littman. Value preserving state-action abstractions. In *International Conference on Artificial Intelligence and Statistics*, pp. 1639–1650, 2020.
- Edward Y. Chang. Trivium: Temporal regret as a first-class objective for causal-memory controllers. Baseline PDF supplied in the project directory; local artifact verifies the probe/log core only.
- Lihong Li, Thomas J. Walsh, and Michael L. Littman. Towards a unified theory of state abstraction for MDPs. In *International Symposium on Artificial Intelligence and Mathematics*, 2006.
- Jiaqi Liu, Yaofeng Su, Peng Xia, Siwei Han, Zeyu Zheng, Cihang Xie, Mingyu Ding, and Huaxiu Yao. Simplemem: Efficient lifelong memory for llm agents. Baseline PDF supplied in the project directory; native LoCoMo results and protocol are reported separately in this paper.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yoav Fang. Evaluating very long-term conversational memory of llm agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pp. 13851–13870, 2024. doi: 10.18653/v1/2024.acl-long.747. URL <https://doi.org/10.18653/v1/2024.acl-long.747>.
- Yijiashun Qi, Xiang Xu, and Yuxuan Li. When not to write memory: Governing false promotion from correlated agent traces. Baseline PDF supplied in the project directory; write-time governance is not silently transferred to access-time authorization.
- Ashish Rana, Chia-Chien Hung, Qumeng Sun, Julian Martin Kunkel, and Carolin Lawrence. Oblivion: Self-adaptive agentic memory control through decay-driven activation. Baseline PDF supplied in the project directory; local verification is limited to the official decay core.
- Baris Simsek. When to forget: A memory governance primitive. Baseline PDF supplied in the project directory; native task and metric are kept separate from the unified QA contract.
- Doyen Wu, Hao Wang, Wenhao Yu, Yiming Zhang, Kai-Wei Chang, and Yiming Dong. Long-memeval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*, 2024. URL <https://arxiv.org/abs/2410.10813>.

Mingshu Zou, Zhen Guo, Lei Liang, Zhihao Wang, Qian Wang, Qiang Wen, Irwin King, Liang Qu, and Zhaocheng Xu. Remember the decision, not the description: A rate-distortion framework for agent memory. *arXiv preprint arXiv:2605.10870*, 2026. URL <https://arxiv.org/abs/2605.10870>. Decision-centric rate-distortion memory with an exact downstream-decision forgetting boundary.

A FORMAL ASSUMPTIONS AND PROOFS

The mathematical scope is as follows. (A1) X_r , score spaces, qualification spaces, and observation spaces are standard Borel; all kernels and payoffs are measurable. (A2) The horizon is finite, $\gamma \in (0, 1]$, terminal payoffs are bounded, and all displayed expectations are integrable. (A3) A score-only commitment policy may randomize from $R_t(x)$ but cannot inspect the within-fiber state. (A4) The closure inequalities in (10) hold simultaneously for every stage from the decision time through H , every relevant state, and every action used in the Bellman recursion. (A5) The quoted radii $\varepsilon_r, \eta_r, M_r$ are fixed before the certificate is evaluated. These are contractual assumptions to be tested by interventions, not properties inferred from a field list or from the public datasets.

A.1 PROOF OF LIFECYCLE ACCOUNTING

By definition, $Q_t(x, a) = \ell_t(x, a) + \gamma C_t^a(x)$. Conditioning first on the next physical state and then on the fresh observation gives

$$\begin{aligned} C_t^a(x) &= \mathbb{E}[V_{t+1}(B^{a,S'}, S') \mid x, a] \\ &\quad + \mathbb{E}[V_{t+1}(B^{a,S',O}, S') - V_{t+1}(B^{a,S'}, S') \mid x, a] \\ &= A_t^a(x) + I_t^a(x). \end{aligned} \tag{18}$$

Substitution proves the action-value identity. Subtracting archive from keep and defining $\Delta_t^{\text{imm}} = \ell_t(\cdot, \text{keep}) - \ell_t(\cdot, \text{archive})$, $\Delta_t^{\text{access}} = A_t^{\text{keep}} - A_t^{\text{archive}}$, and $\Delta_t^{\text{info}} = I_t^{\text{keep}} - I_t^{\text{archive}}$ proves the contrast decomposition.

A.2 PROOF OF THEOREM 1

Let p be the probability that the score-only rule chooses keep on the common score value in (6). In x_+ , choosing archive has regret at least m_+ , so the expected regret is at least $(1-p)m_+$. In x_- , choosing keep has regret at least m_- , so it is at least pm_- . Balancing the two lower bounds gives

$$\max\{(1-p)m_+, pm_-\} \geq \frac{m_+m_-}{m_+ + m_-}, \tag{19}$$

which proves the claim. No tie convention is needed because both margins are strict.

A.3 PROOF OF THEOREM 2

We prove the first bound in (12) by backward induction. At $H+1$, it is exactly the terminal inequality in (10). Suppose it holds at $r+1$. Let $P_x = (q_{r+1})_{\#} T_r^a(\cdot \mid x)$. The Bellman recursions, the induction hypothesis, and the definition of total variation give

$$|Q_r(x, a) - \bar{Q}_r(q_r(x), a)| \leq \varepsilon_r + \gamma \left| \int V_{r+1} dT_r^a - \int \bar{V}_{r+1} d\bar{T}_r^a \right| \tag{20}$$

$$\leq \varepsilon_r + \gamma \left(b_{r+1} + \left| \int \bar{V}_{r+1} dP_x - \int \bar{V}_{r+1} d\bar{T}_r^a \right| \right) \tag{21}$$

$$\leq \varepsilon_r + \gamma (b_{r+1} + 2M_{r+1}\eta_r) = b_r. \tag{22}$$

Maximization preserves the same bound for V_r and $\bar{V}_r$, completing the induction. Applying the action-value bound once for keep and once for archive yields the contrast bound. If $\bar{\Delta}_r > 2b_r$, then $\Delta_r > 0$; if $\bar{\Delta}_r < -2b_r$, then $\Delta_r < 0$. Thus each committed action in (13) has the full-state optimal sign. The theorem makes no assertion in the remaining band.

Each result row is tied to a dataset hash, command, configuration, scorer hash, model/API version, seed, failure count, and output hash. The v3 inferential unit is the unique world signature; the nine mechanism families are the bootstrap clusters. Public metrics use frozen chronological masking and the official LoCoMo scorer where commensurate. Authorized and physical storage are recorded separately. Missing metrics remain missing.

B FROZEN READER

For a sentence s and question q , let $\mathcal{W}$ be a fixed stopword set, n the number of exposed sentences, and $\text{df}(w)$ the number of those sentences containing w :

$$h(s, q) = \sum_{w \in (q \cap s) \setminus \mathcal{W}} \left[1 + \log \frac{1 + n}{1 + \text{df}(w)} \right] + 1.25 \mathbf{1}\{\text{type}(q, s)\} - 0.65 \mathbf{1}\{\text{ack}(s)\} - \gamma \mathbf{1}\{\text{echo}(s)\}. \quad (23)$$

The type indicator rewards explicit date/time forms for temporal questions and numeric forms for count questions; *ack* flags generic acknowledgements (e.g., “thanks” or “wow”). The reader returns the highest-scoring packed sentence, with deterministic ties by exposure order. For the original shared readout $\gamma = 0$; the structured response-prior variant and its BM25 control set $\gamma = 0.85$, where *echo* flags a sentence beginning with an interrogative form. This penalty exists because a retrieved question can otherwise outscore the declarative sentence that answers it.

C CLAIM–EVIDENCE BOUNDARY

Table 7: Claim–evidence map used to constrain the narrative. The third column is the part we ask reviewers to hold us to: each entry names a reading the evidence does not support.

Claim	Direct evidence	What it does <i>not</i> claim
A relevance score can be provably insufficient	Theorem 1; six of eight published surfaces have opposite-sign fibers (Table 1)	The audit covers transported <i>scores</i> , not full systems; margins are those of our declared objective
Qualification state suffices under audited closure	Theorem 2 propagated radius and signed commitment guarantee	An analytic condition; ε_r, η_r are not estimated on real traces here
Abstention recovers near-oracle value	98.9% of oracle value at lower authorized storage; never worse than SimpleMem on 225/225 (Table 4)	Value CIs cross zero at 9 family clusters; <i>archive_all</i> is cheaper if missed commits are free
Each field is load-bearing	Family-localized ablations, 74.447 regret from lineage alone (Table 5)	Under the declared generator; <i>scope_mismatch</i> stays unresolved for all gold-free policies
Authorization cost can fall without losing retrieval	97.6%/97.8% storage reduction at higher hit/recall than full retention	No QA superiority; absolute F1 is format-sensitive and we do not lead on it

D ADDITIONAL LIFECYCLE DIAGNOSTICS

Access-layer ablation anchor. Base SQCAD without the FIR filter or the response-edge prior authorizes 329.9 tokens on LoCoMo at hit/recall 0.2122/0.1899 (reader context 348.2, answer F1 0.0341 Qwen / 0.0204 Terra), and 1,631.2 tokens on LongMemEval-S at 0.7542/0.1149 (reader context 2,721.7, token F1 0.2528 / 0.2857). Both access variants therefore improve exposure at approximately unchanged authorized storage, which is the intended separation: the guard changes what the reader sees, not what remains permitted.

Score-fiber duplication. Four surfaces in Table 1 share the same widest witness because their scores collide on the same *version_update* pair. We report them as separate rows because each is a separate published score, but they are not four independent pieces of evidence; the distinct witnesses are those of FadeMem, GovMem, and the shared *version_update* pair.

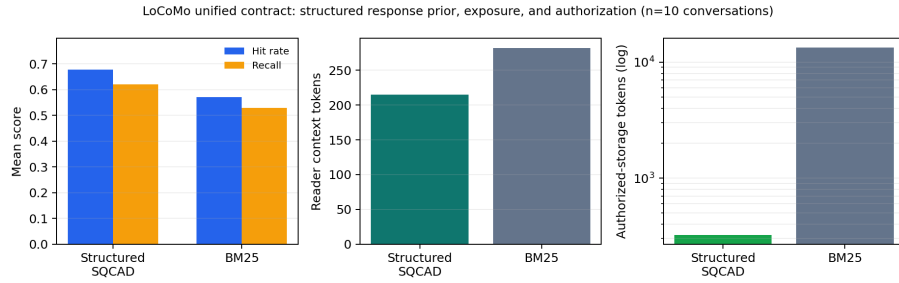


Figure 2: LoCoMo quality–context–authorized-storage comparison under the frozen shared-reader contract, plotting the frontier tabulated in Table 2.

Statistical detail. Family-cluster bootstrap uses 4,000 resamples over the 9 mechanism families with a fixed seed, resampling whole families rather than worlds. Sign tests are exact one-sided binomial tests over family means with ties dropped. Per-world dominance counts are reported directly since they need no distributional assumption. Full component ablations, cost-normalized utility, and storage/probe sensitivity are supplied with the artifact manifest.

E NAMED-BASELINE AUDIT

Table 8: Reproduction ledger. Native values are transcribed from supplied papers/PDFs and are not numerically mixed with unified-contract rows. MF denotes a mechanism-faithful implementation; R-C denotes an open-weight or paper-mechanism substitute.

Baseline	Native task/metric	Local status and boundary	Tier
DeMem	Decision rate–distortion	Literature comparator; no local native row in this freeze	R-A
SimpleMem	LoCoMo F1/BLEU/cost	Qwen3-8B open-weight substitute; unified hit 0.7051, recall 0.6524, storage 19,642.8; no native delta	R-C
ActMem	LoCoMo QA accuracy	Qwen3-8B/Qwen3-Embedding MF substitute; unified hit 0.6317, recall 0.5729, storage 10,528; native accuracy not reproduced	MF
Oblivion	LongMemEval-S	Official decay core verified; end-to-end path open	R-D
Memory Worth	Spearman ρ	Synthetic MF calibration $\rho = 0.8977$ at 10k over 20 seeds; text retrieval open	MF
Trivium	Temporal regret	Probe/log core only; native end-to-end path open	MF
GovMem	Promotion precision/recall	Controlled simulator and noise sweep; not GovMem-Bench reproduction	R-D/E